

A Technical Introduction to Named entity recognition

Section 1

Precision is the number of predicted entity name spans that line up exactly with spans in the gold standard evaluation data. I.e. when [Person Hans] [Person Blick] is predicted but [Person Hans Blick] was required, precision for the predicted name is zero. Precision is then averaged over all predicted entity names. Recall is similarly the number of names in the gold standard that appear at exactly the same location in the predictions. F1 score is the harmonic mean of these two. It follows from the above definition that any prediction that misses a single token, includes a spurious token, or has the wrong class, is a hard error and does not contribute positively to either precision or recall. Thus, this measure may be said to be pessimistic: it can be the case that many "errors" are close to correct, and might be adequate for a given purpose. For example, one system might always omit titles such as "Ms." or "Ph.D.", but be compared to a system or ground-truth data that expects titles to be included. In that case, every such name is treated as an error. Because of such issues, it is important actually to examine the kinds of errors, and decide how important they are given one's goals and requirements. Evaluation models based on a token-by-token matching have been proposed. Such models may be given partial credit for overlapping matches (such as using the Intersection over Union criterion). They allow a finer grained evaluation and comparison of extraction systems.

Section 2

Formal evaluation To evaluate the quality of an NER system's output, several measures have been defined. The usual measures are called precision, recall, and F1 score. However, several issues remain in just how to calculate those values. These statistical measures work reasonably well for the obvious cases of finding or missing a real entity exactly; and for finding a non-entity. However, NER can fail in many other ways, many of which are arguably "partially correct", and should not be counted as complete success or failures. For example, identifying a real entity, but:

Section 3

Early work in NER systems in the 1990s was aimed primarily at extraction from journalistic articles. Attention then turned to processing of military dispatches and reports. Later stages of the automatic content extraction (ACE) evaluation also included several types of informal text styles, such as weblogs and text transcripts from conversational telephone speech conversations. Since about 1998, there has been a great deal of interest in entity identification in the molecular biology, bioinformatics, and medical natural language processing communities. The

most common entity of interest in that domain has been names of genes and gene products. There has been also considerable interest in the recognition of chemical entities and drugs in the context of the CHEMDNER competition, with 27 teams participating in this task. In 2001, research indicated that even state-of-the-art NER systems were brittle, meaning that NER systems developed for one domain did not typically perform well on other domains. Considerable effort is involved in tuning NER systems to perform well in a new domain; this is true for both rule-based and trainable statistical systems. As of 2007, state-of-the-art NER systems for English produce near-human performance. For example, the best system entering MUC-7 scored 93.39% of F-measure while human annotators scored 97.60% and 96.95%.

Section 4

Another field that has seen progress but remains challenging is the application of NER to Twitter and other microblogs, considered "noisy" due to non-standard orthography, shortness and informality of texts. NER challenges in English Tweets have been organized by research communities to compare performances of various approaches, such as bidirectional LSTMs, Learning-to-Search, or CRFs.

Section 5

Jurafsky, Daniel; Martin, James H. (2008). "13.5 Partial Parsing". *Speech and Language Processing* (2nd ed.). Upper Saddle River, N.J.: Prentice Hall. ISBN 978-0131873216. Jurafsky, Daniel; Martin, James H. (2008). "22.1. Named Entity Recognition". *Speech and Language Processing* (2nd ed.). Upper Saddle River, N.J.: Prentice Hall. ISBN 978-0131873216.

Section 6

Current challenges Despite high F1 numbers reported on the MUC-7 dataset, the problem of named-entity recognition is far from being solved. The main efforts are directed to reducing the annotations labor by employing semi-supervised learning, robust performance across domains and scaling up to fine-grained entity types. In recent years, many projects have turned to crowdsourcing, which is a promising solution to obtain high-quality aggregate human judgments for supervised and semi-supervised machine learning approaches to NER. Another challenging task is devising models to deal with linguistically complex contexts such as Twitter and search queries. There are some researchers who did some comparisons about the NER performances from different statistical models such as HMM (hidden Markov model), ME (maximum entropy), and CRF (conditional random fields), and feature sets. And some researchers recently proposed graph-based semi-supervised learning model for language specific NER tasks. A recently emerging task of

identifying "important expressions" in text and cross-linking them to Wikipedia can be seen as an instance of extremely fine-grained named-entity recognition, where the types are the actual Wikipedia pages describing the (potentially ambiguous) concepts. Below is an example output of a Wikification system:

Section 7

[Jim]Person bought 300 shares of [Acme Corp.]Organization in [2006]Time. In this example, a person name consisting of one token, a two-token company name and a temporal expression have been detected and classified.

Section 8

Lexical items: The token itself be labeled. Stemmed lexical items. Shape: The orthographic pattern of the target word. For example, all lowercase, all uppercase, initial uppercase, mixed case, uppercase followed by a period (often indicating a middle name), contains hyphen, etc. Affixes of the target word and surrounding words. Part of speech of the word. Whether the word appears in one or more named entity lists (gazetteers). Words and/or n-grams occurring in the surrounding context. A gazetteer is a list of names and their types, such as "General Electric". It can be used to augment any system for NER. They had been often used in the era of statistical machine learning. Many different classifier types have been used to perform machine-learned NER, with conditional random fields being a typical choice. Transformers features token classification using deep learning models.